B2-PROPA-HLAYER claim-level synthesis: Proposition A and the discharge of H-LAYER

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

Claim-level synthesis for B2-PROPA-HLAYER. It adds no mathematics; every statement carries the unit's registered tier. **Scope:** the composition of B2's units into the canonical claim state. **Not claimed:** anything beyond the enacted scope; in particular Prop-A's own bundle scope is $\mathcal{A}_{\mathrm{adm}}$ ($T' \leq 13$) and the C_{full} statement runs through the joint B1/B2 head package, not through Prop-A alone.

2. The claim and its canonical state

Claim (canonical). The isotropic Gaussian–Hartree layer is the strict comparison infimum: the H-LAYER hypothesis class is DISCHARGED – every analytic H-LAYER residual is closed class-wide on the primary admissible class, and the layer comparison enters the Reading-H selection as a theorem, not an assumption. Tier: T7-SCOPE$_{C_{\mathrm{full}}}$ (jointly with B1, enacted 2026-06-10); hypothesis: A1-KERNEL-CONV; open gates: none.

3. Sub-proof unit map (per-unit tier and role)

Unit	Tier	What it contributes	Publication state
`Prop-A/`	T6	(D) block-diagonal isotropy (entropy floor $\frac{1}{2}r_R^2 > 0$, breathing $\frac{3}{2}u_{\mathrm{eff}} > 0$) + (O) class-wide closure on $\mathcal{A}_{\mathrm{adm}}$ ($R_{\mathrm{lead}}(13) = 0.650 < 1$, joint $= 1.097 > 1$, $T'_{\mathrm{crit}} = 60.4$); T-016..T-024 chain	PUBLISHED – `Prop-A-T6-260611` (9/9 scripts, 44/44)
`G-A0-DUI/`	T6/T7-comp.	closed lemma: $M'(m) < 0$ by differentiation under the integral (dominated convergence) – supports $A = 0$ uniqueness of the reference	AUXILIARY (operator-accepted v1.1; candidate fold into Prop-A/A1)
`H-A0-removal/`	T6/T7-comp.	sign-decomposition theorem $F_0' = \frac{1}{2}M'g$ discharging H-A0's U, Z to H-ANCHOR within the certified window (zero-at-gap anchor $g(m_{\mathrm{gap}}) = 0$)	AUXILIARY (operator-accepted v1.1; candidate fold into Prop-A/A1)

External: the joint B1/B2 head package `Reading-H-cFull-T7-260611` carries the C_{full} extension; Prop-A is its (D)/(O) primary-class sub-structure ($\mathcal{A}_{\mathrm{adm}} \subset \mathcal{C}_{\mathrm{full}}$).

4. Composition and what remains

Prop-A (T6, PUBLISHED) closes every analytic H-LAYER residual class-wide on $\mathcal{A}_{\mathrm{adm}}$ at the operating point via the three sectors (D)/(O)/(S); the two $A = 0$ lemmas secure the reference-state uniqueness feeding the A1 convention; the C_{full} statement is enacted at the head. Nothing in B2's gate list remains open. Remaining (non-blocking, registered): the operator fold-in decision for `G-A0-DUI/H-A0-removal` into Prop-A or the A1 record (cosmetic restructuring); the A1-KERNEL-CONV migration itself (claim-A1-side, per main-proof-line.md).

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"B2 is T7-SCOPE but its only published in-claim bundle is T6 – mismatch?"* DISMISSED with explanation: the T7-SCOPE state is the JOINT B1/B2 head enactment (Reading-H package); Prop-A's T6 is its primary-class sub-structure and was deliberately NOT re-tiered upward (operator ruling 2026-06-11: 'T6 NOT T7, does not independently enact full C_full'). The claim-level tier and the unit-level tier answer different questions; both are quoted.

(β) *"The $A = 0$ lemmas might be load-bearing yet only operator-accepted drafts."* VALID with mitigation: they support the A1 REFERENCE definition (input layer), not the comparison chain; A1 itself is the registered hypothesis of both B1 and B2, so their weight is carried by the hypothesis declaration, not silently.

(γ) *"T-019 history (hdiag_gershgorin_rowsum exchange-sign framing) was superseded – does the synthesis cite stale framing?"* DISMISSED: the superseded framing was replaced by the res5_019 local-functional reframing ($f''(M) = \frac{3}{2}u_{\text{eff}} > 0$) and the bundle carries the corrected note (Prop-A v1.3, T-019 supersession note retained as history).

Quantitative sanity check (mandatory). The quoted Prop-A load-bearing values ($\Phi''_{\text{diag}} = 4.0275 > 0$, $\frac{1}{2}r_R^2 = 4.6368 \times 10^{-2} > 0$, $R_{\text{lead}}(13) = 0.650 < 1$, $\text{joint}(13) = 1.097 > 1$, $T'_{\text{crit}} = 60.4$) are the operator-confirmed bundle values (9/9, 44/44); margins 0.350 and 0.097 are positive; the $\mathcal{A}_{\text{adm}} \subset \mathcal{C}_{\text{full}}$ inclusion is definitional ($T' \leq 13$ vs the class cap).

6. Result footer

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Result ID:               B2-PROPA-HLAYER-Synthesis-T013-260612
                         (PUBLISHED claim-level SYNTHESIS, T-013)
Precise statement:       B2's canonical state (T7-SCOPE_{C_full} given
                         A1; H-LAYER discharged; gates none) is carried
                         by the joint B1/B2 head package plus the
                         PUBLISHED Prop-A-T6-260611 primary-class
                         sub-structure; G-A0-DUI and H-A0-removal are
                         auxiliary A=0-reference lemmas feeding A1.
Scope:                   synthesis layer; no new math, no tier change,
                         no gate flip.
Evidence grade:          SYNTHESIS over PUBLISHED + operator-accepted
                         units.
Reproduction command:    run the Prop-A bundle entry manifest (9
                         scripts); head package per its bundle.
Falsifier:               any falsifier of the cited footers; or a
                         unit-tier mismatch against INDEX.md.
Tier before / after:     no change (B2 stays T7-SCOPE_{C_full}; Prop-A
                         unit stays T6).
No-overclaim statement:  Prop-A alone does not enact C_full; the
                         claim-level tier is the joint head enactment.
Next required action:    operator: optional fold-in decision for the
                         two A=0 lemmas (cosmetic restructuring).
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